

NY Reduced Power-System Model:

Current Progress

Validation of a 49-bus research model for interface-flow and preliminary dynamic line-rating studies

NPCC / NY Reduced-Model Workstream

Status through July 14, 2026 | Prepared July 19, 2026

A self-contained update for technical audiences new to the project

Executive Summary

- ▶ A **49-bus New York reduced model is now accepted** for the summer-peak operating point and two reactive-load sensitivity tests.
- ▶ The accepted model passes AC power flow, generator and voltage limits, trusted branch limits, interface-flow accuracy, dispatch-movement limits, and a full-to-reduced structural benchmark.
- ▶ Four transfer interfaces are calibrated, and five physical transmission circuits have current results that can support later conductor mapping.
- ▶ **Winter peak and high-Total-East conditions remain excluded** because they exceed fixed dispatch and/or interface-accuracy limits.

ACCEPTED

Summer-peak research case

71.7 MW

Calibrated-interface MAE

5 circuits

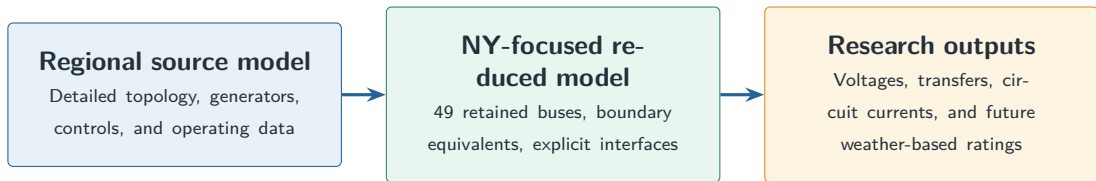
Physical current scope

Claim boundary

The model supports preliminary dynamic line-rating methodology research. It does not yet contain the physical and operational data needed for utility planning or real-time operating claims.

Source: Current validation, interface, dispatch, and structural benchmark ledgers

What Are We Building?



Reduced model

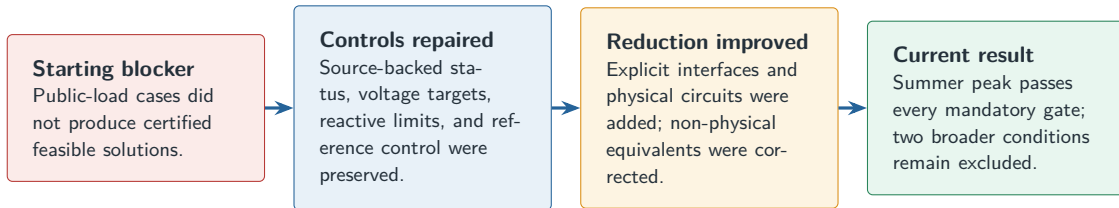
A smaller electrical equivalent that preserves the behavior needed for selected New York studies while removing most of the surrounding regional detail.

Dynamic line rating

A weather-dependent current limit. This deck validates electrical currents; conductor and weather inputs are a later layer.

Source: Current model architecture and validation documentation

How the Project Moved Forward



The major process change

Cases are no longer accepted because they merely converge. Publication now requires electrical feasibility, realistic control provenance, bounded redispatch, interface accuracy, and full-to-reduced structural agreement.

Source: Project handoff and current validation policy

Current Model Configuration

Network and operating point

- ▶ 49 buses, 83 branches, 49 generator records
- ▶ New York zones A–K plus external boundary equivalents
- ▶ Five explicitly mapped physical circuits
- ▶ Summer load: 30,644.99 MW
- ▶ Generation: 31,433.46 MW
- ▶ Active loss: 788.47 MW

Controls and measurements

- ▶ 35 regulating generators at seven eligible buses
- ▶ Source-backed active and reactive limits
- ▶ Fixed external active-power schedules
- ▶ Fixed-zero external reactive schedule by default
- ▶ Four calibrated interface operators
- ▶ Three additional interface proxies retained as diagnostics

Why the circuit count increased

Two retained corridors are represented as separate parallel circuits, producing 83 model branches rather than 81 while preserving five one-to-one physical circuit rows.

Source: Current case dimensions, control map, and dispatch/loss ledger

What the Validation Gates Mean

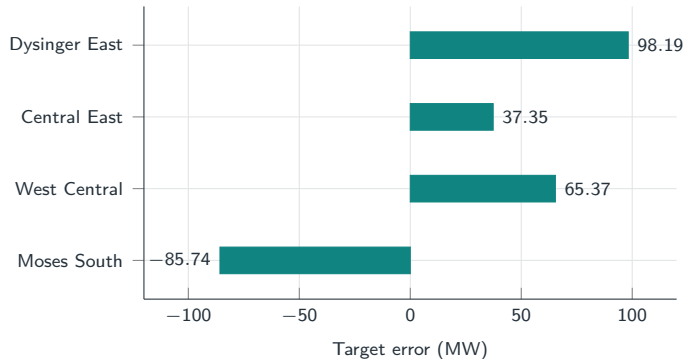
Gate family	Requirement	Current result	Status
Power flow	Standard and reactive-limit AC power flow	both converge	PASS
Restoration	No artificial P/Q/voltage/branch slack	0	PASS
Physical limits	Voltage, generator P/Q, trusted branch violations	0 / 0 / 0 / 0	PASS
Power balance	Reference adjustment below 1 MW	1.17×10^{-6} MW	PASS
Interfaces	$\text{MAE} \leq 100$; $\text{max} \leq 200$; $ \text{bias} \leq 50$ MW	71.7 / 98.2 / 28.8	PASS
Dispatch	Movement $\leq 10\%$ and ≤ 3000 MW	6.433%; 1,971 MW	PASS
Controls	No unexpected regulating buses; reference matched	0; matched	PASS
Structural match	Current, loss, and net-Q errors each $\leq 15\%$	all below 15%	PASS

Margin indicators

Trusted static maximum loading is 56.3% of its stated rating, and no regulating group is at a reactive limit. Voltages reach approximately 0.90–1.10 pu, so hard bounds pass but preferred voltage margin remains narrow.

Source: Current mandatory and preferred validation ledger

Accepted Summer-Peak Interface Results



71.7 MW

Mean absolute error

98.2 MW

Maximum error

28.8 MW

Signed bias

All aggregate gates pass: $MAE \leq 100$,
maximum ≤ 200 , and $|bias| \leq 50$ MW.

Source: Current summer-peak interface validation ledger; calibrated operators only

Operating Conditions Tested

Condition	Reactive load	IF MAE MW	Max IF MW	Dispatch MW	Load %	Decision
Summer peak	Nominal	71.7	98.2	1,971	6.433	Accepted
Summer peak	90%	71.5	98.1	1,971	6.433	Accepted
Summer peak	110%	71.6	98.1	1,971	6.433	Accepted
Winter peak	All tests	41.4	80.9	3,199	13.602	Excluded
High Total East	All tests	156.8	318.4	3,410	12.037	Excluded

Winter peak

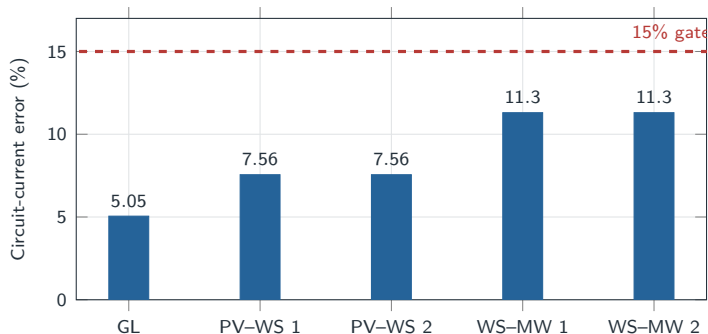
Electrical and interface tests pass, but the case requires more than 3,000 MW of zonal redispatch and more than 10% of total load.

High Total East

The case exceeds both redispatch limits and the calibrated-interface accuracy limits.

Source: Current dispatch, loss, and interface validation ledgers

Full-to-Reduced Structural Benchmark



0.000096%

Active-loss error

2.470%

Net-network-Q error

PASS

Reactive-limit power flow

What this demonstrates

For the same 2019 operating snapshot, the reduced network matches selected physical-circuit currents, active losses, and net reactive behavior within the adopted 15% research threshold.

Dynamic-Line-Rating Circuit Scope

Physical corridor	Circuit count	Summer current
Gilboa–Leeds	1	1.157 kA
Pleasant Valley–Wood Street	2	0.925 kA each
Wood Street–Millwood	2	0.591 kA each

What is usable now

The five mapped rows carry terminal and series currents with physical-circuit provenance suitable for later conductor assignment.

What is not yet included

- ▶ Conductor type and ampacity
- ▶ Thermal parameters and weather
- ▶ Route segmentation and sag/clearance
- ▶ Terminal-equipment ratings
- ▶ Dynamic ampacity calculation

Interpretation rule

Currents on aggregate and boundary equivalents are electrical diagnostics only. A model branch rating is not automatically a conductor ampacity or dynamic rating.

Source: Current branch-current and physical-circuit mapping ledgers

Remaining Issues and Limitations

Operating coverage

- ▶ Winter peak and high-Total-East conditions fail fixed acceptance gates.
- ▶ Total East, UPNY–ConEd, and Dunwoodie South remain diagnostic because their complete monitored-circuit definitions are not conserved.
- ▶ Summer voltages reach approximately 0.90/1.10 pu: hard bounds pass, but preferred margin is narrow.

Reduction and physical fidelity

- ▶ The non-physical equivalent correction is empirical rather than an exact Ward reduction.
- ▶ Three control locations remain provisional fixed-reactive-power buses.
- ▶ One legacy equivalent rating overload remains diagnostic because its rating meaning is not trusted.
- ▶ No conductor or weather layer has been attached.

Claim boundary

The model is a defensible preliminary research equivalent, not a utility planning, reliability, facility-rating, real-time dispatch, or operating model.

Source: Current limitations and mandatory-gate documentation

Recommended Next Work

1. **Complete interface definitions:** replace diagnostic proxies with explicit conserved circuit or cutset operators where the retained topology supports them.
2. **Broaden operating coverage without relaxing gates:** reduce winter and high-Total-East redispatch, and improve high-Total-East interface closure.
3. **Increase voltage-control margin:** investigate buses near 0.90/1.10 pu and resolve the three provisional control locations without placeholder reactive limits.
4. **Add the physical rating layer:** map conductors, equipment limits, route segments, weather, thermal models, and sag/clearance constraints to the five circuits.
5. **Revalidate before publication:** rerun all operating and structural gates and publish only cases that pass every mandatory test.

Near-term deliverable

An updated validation package that preserves the accepted summer-peak model and either passes the broader operating conditions under unchanged gates or keeps them explicitly excluded.

Current takeaway

Use the accepted summer-peak model for preliminary dynamic line-rating methodology development.

Keep winter peak and high-Total-East conditions excluded. Do not make utility-operational or dynamic-rating claims until interface coverage and physical conductor/weather inputs are complete.

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